

Name \_\_\_\_\_ Teacher \_\_\_\_\_



## **GOSFORD HIGH SCHOOL**

**2014**

### **HIGHER SCHOOL CERTIFICATE**

### **ASSESSMENT TASK 3**

# **MATHEMATICS – EXTENSION 1**

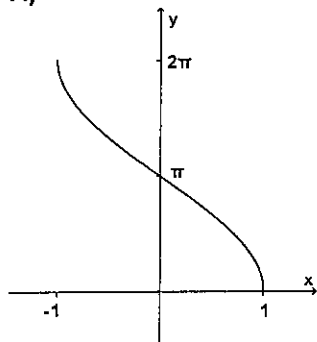
**Duration-** 60 minutes plus 5 minutes reading time

|                      |                                                                                 |            |
|----------------------|---------------------------------------------------------------------------------|------------|
| <b>Questions 1-5</b> | Multiple choice. Answer on the <b>multiple choice response sheet provided</b> ) | <b>/5</b>  |
| <b>Question 6</b>    | Inequalities, Division of a line, Iterative methods.                            | <b>/7</b>  |
| <b>Question 7</b>    | Inverse Functions, Inverse Trigonometric functions                              | <b>/11</b> |
| <b>Question 8</b>    | Inverse Trigonometric Functions                                                 | <b>/11</b> |
| <b>Question 9</b>    | Binomial Theorem                                                                | <b>/11</b> |
| <b>TOTAL</b>         |                                                                                 | <b>/45</b> |

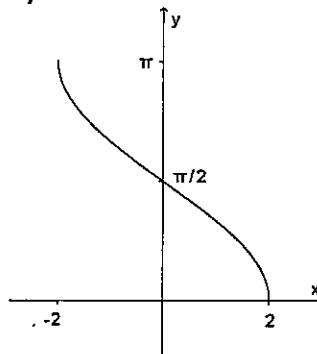
Questions 1-5. 1 mark each. Answer on the multiple choice answer sheet provided.

Question 1. The best representation of the graph of  $y = \cos^{-1} 2x$  is given by:

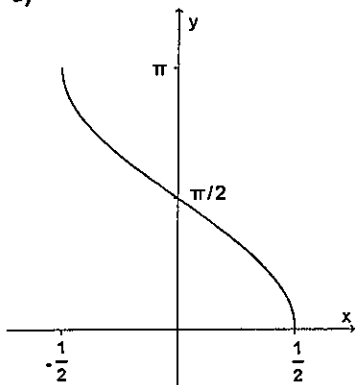
A)



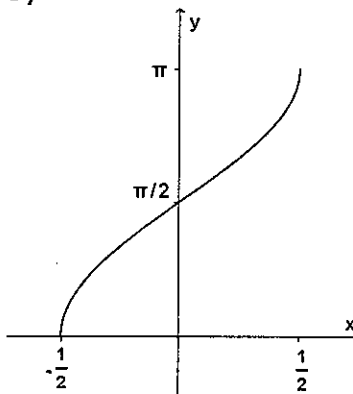
B)



C)



D)



Question 2. The coefficient of the third term in the expansion of  $(3 - 2y)^8$  is:

A) -81648

B) 81648

C) -108864

D) 16128

Question 3.  $\int \frac{2}{16+x^2} dx =$

A)  $\frac{1}{4} \tan^{-1} \left( \frac{x}{4} \right) + c$

B)  $\frac{1}{2} \tan^{-1} \left( \frac{x}{4} \right) + c$

C)  $2 \tan^{-1} \left( \frac{x}{4} \right) + c$

D)  $\frac{1}{8} \tan^{-1} \left( \frac{x}{16} \right) + c$

Question 4. Given that  $a$  is an approximation to a root of the equation  $y = \sin^{-1}x$ , then a better approximation using Newton's method is given by:

- A)  $a - \frac{\sin^{-1}a}{\sqrt{1-a^2}}$       B)  $a + \sin^{-1}a \sqrt{1-a^2}$
- C)  $a - \sin^{-1}a \sqrt{1-a^2}$       D)  $a - \frac{1}{\sin^{-1}a \sqrt{1-a^2}}$

Question 5. Which of the following relationships is true?

- A)  ${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$       B)  ${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_r$
- C)  ${}^nC_r - {}^nC_{r+1} = {}^{n+1}C_{r+1}$       D)  ${}^nC_r \times {}^nC_{r+1} = {}^{n+1}C_{r+1}$

Question 6. **START A NEW PAGE.**

- a) Solve  $\frac{x^2-16}{x} \geq 0$  (3)
- b) The point  $Q$  divides the interval joining  $R(6, -3)$  and  $P(-2, -5)$  externally in the ratio 3:2. Find the coordinates of  $Q$ . (2)
- c) The equation  $e^x = 4x + 8$  has a root close to  $x = 3$ . Using 3 as the first approximation and Newton's method once, find an improved value of this root. Give answer correct to 3 decimal places. (2)

Question 7. **START A NEW PAGE.**

a) i) Show that  $\frac{x+2}{x+1} = 1 + \frac{1}{x+1}$  (1)

ii) Carefully sketch the function defined by

$$g(x) = \frac{x+2}{x+1} \text{ for } -1 < x$$
 (3)

iii) Find  $g^{-1}(x)$  and sketch it on the same graph as  $g(x)$ . (3)

iv) Find any value(s) of  $x$  for which  $g(x) = g^{-1}(x)$  (2)

b) Find the exact value of  $\sin \left[ 2 \tan^{-1} \left( \frac{3}{4} \right) \right]$  (2)

Question 8. **START A NEW PAGE.**

a) Differentiate  $\sin^{-1} 4x$  (1)

b) Find  $\int \frac{x+1}{x^2+4} dx$  (2)

c) Evaluate  $\int_0^2 \frac{1}{\sqrt{16-x^2}} dx$  (2)

d) The curve  $y = \frac{1}{\sqrt{1+x^2}}$  is rotated about the  $x$  axis. Find the volume of the solid enclosed between  $x = \frac{1}{\sqrt{3}}$  and  $x = \sqrt{3}$ . (3)

e) Differentiate  $x \sin^{-1} x$  and hence find  $\int_0^1 \sin^{-1} x dx$  (3)

Question 9. **START A NEW PAGE.**

- a) Using the fact that  $(1+x)^4(1+x)^9 = (1+x)^{13}$ , show that:

$${}^4C_0 {}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + {}^4C_4 {}^9C_0 = {}^{13}C_4 \quad (3)$$

- b) By assigning an appropriate value to  $x$  in the expansion of  $(1+x)^n$ , prove

$${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_{n-1} = 2^n - 2 \quad (2)$$

- c) Consider the binomial expansion

$$1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n$$

Show that:

$$\text{i) } 1 - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0 \quad (1)$$

$$\text{ii) } \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} = n(2)^{n-1} \quad (2)$$

$$\text{iii) } 1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + (-1)^n \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1} \quad (3)$$

# Solutions

1) C

C  
B  
B  
C  
A

$$2) {}^8C_2 3^6 (-2)^2$$

$$= 81648$$

3) B

$$4) a_1 = a - \frac{\sin^{-1} a}{\sqrt{1-a^2}}$$

$$= a - \sqrt{1-a^2} \cdot \sin^{-1} a$$

$\therefore$  C.

5) A

Q 6

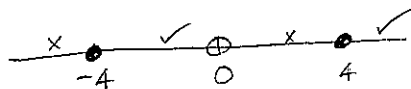
$$2) \frac{x^2-16}{x} \geq 0$$

critical point  $x \neq 0$

$$\text{Solve } \frac{x^2-16}{x} = 0$$

$$x^2 = 16$$

$$x = \pm 4$$



Test:  $x = -5$  F

$x = -1$  T

$x = 1$  F

$x = 5$  T

$$-4 \leq x < 0 \text{ or } x \geq 4$$

$$b) K(6, -3) \quad P(-2, -5)$$

$$3: -2$$

$$\left( \frac{-12-6}{1}, \frac{6-15}{1} \right)$$

$$= (-18, -9)$$

$$c) e^x = 4x + 8$$

$$e^x - 4x - 8 = 0$$

$$f(3) = e^3 - 12 - 8$$

$$= 0.0855$$

$$f'(x) = e^x - 4$$

$$f'(3) = e^3 - 4$$

$$= 16.0855$$

$$a_1 = a - \frac{f(a)}{f'(a)}$$

$$= 3 - \frac{0.0855}{16.0855}$$

$$= 2.995.$$

Question

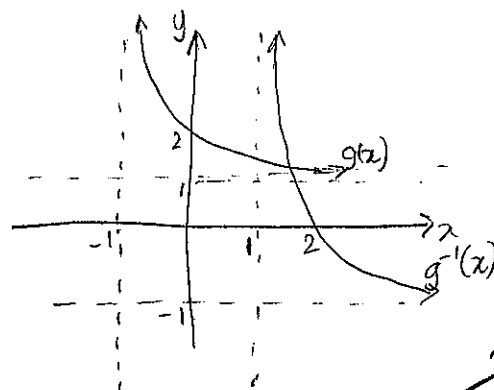
## Question 7

$$a) \quad i) \quad \frac{x+2}{x+1} = \frac{x+1+1}{x+1}$$

$$= \frac{x+1}{x+1} + \frac{1}{x+1}$$

$$= 1 + \frac{1}{x+1}$$

$$ii) \quad g(x) = \frac{x+2}{x+1}$$



$$iii) \quad x = \frac{y+2}{y+1}$$

$$xy + x = y + 2$$

$$xy - y = 2 - x$$

$$y(x-1) = 2-x$$

$$y = \frac{2-x}{x-1}$$

$$= \frac{1}{x-1} - \frac{(x-1)}{x-1} = \frac{1}{x-1} - 1$$

$$iv) \quad g(x) = g^{-1}(x) \text{ when } y = x.$$

$$\therefore \frac{2-x}{x-1} = x$$

$$2-x = x^2 - x$$

$$x^2 = 2$$

$$x = \sqrt{2}$$

( $-\sqrt{2}$  outside domain)

$$g: y = 1 + \frac{1}{x+1}$$

$$g^{-1}: x = 1 + \frac{1}{y+1}$$

$$x-1 = \frac{1}{y+1}$$

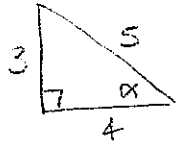
$$y+1 = \frac{1}{x-1}$$

$$y = -1 + \frac{1}{x-1}$$

$$b) \sin \left[ 2 \tan^{-1} \left( \frac{3}{4} \right) \right]$$

$$\tan^{-1} \left( \frac{3}{4} \right) = \alpha$$

$$\tan \alpha = \frac{3}{4}$$



$$\begin{aligned} \sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ &= 2 \cdot \frac{3}{5} \cdot \frac{4}{5} \\ &= \frac{24}{25} \end{aligned}$$

Q8

$$\begin{aligned} a) \frac{d}{dx} \sin^{-1} 4x \\ &= \frac{4}{\sqrt{1-16x^2}} \end{aligned}$$

$$\begin{aligned} b) \int \frac{x+1}{x^2+4} dx \\ &= \int \frac{x}{x^2+4} dx + \int \frac{1}{x^2+4} dx \\ &= \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} + C \end{aligned}$$

$$c) \int_0^2 \frac{1}{\sqrt{16-x^2}} dx$$

$$\begin{aligned} &= \left[ \sin^{-1} \frac{x}{4} \right]_0^2 \\ &= \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{6} \end{aligned}$$

$$d) y = \frac{1}{\sqrt{1+x^2}}$$

$$\begin{aligned} V &= \pi \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{1}{1+x^2} dx \\ &= \pi \left[ \tan^{-1} x \right]_{1/\sqrt{3}}^{\sqrt{3}} \\ &= \pi \left[ \tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right] \\ &= \pi \left[ \frac{\pi}{3} - \frac{\pi}{6} \right] \\ &= \frac{\pi^2}{6} \text{ cubic units} \end{aligned}$$

$$e) \frac{d}{dx} x \sin^{-1} x = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$\therefore \sin^{-1} x = \frac{d}{dx} x \sin^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$\begin{aligned}
 \int_0^1 \sin^{-1} x \, dx &= \int_0^1 \frac{d}{dx} (x \sin^{-1} x) \, dx - \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx \\
 &= [x \sin^{-1} x]_0^1 - \frac{1}{2} \int_0^1 2x(1-x^2)^{-1/2} \, dx \\
 &= \sin^{-1} 1 + \left[ (1-x^2)^{1/2} \right]_0^1 \\
 &= \frac{\pi}{2} + [0 - 1] \\
 &= \frac{\pi}{2} - 1
 \end{aligned}$$

### Question 9

a)  ${}^{13}C_4$  is the coefficient of  $x^4$  in  $(1+x)^{13}$

in  $(1+x)^4(1+x)^9$ , the term in  $x^4$  is:

$${}^4C_0 \cdot {}^9C_4 x^4 + {}^4C_1 x {}^9C_3 x^3 + {}^4C_2 x^2 {}^9C_2 x^2 + {}^4C_3 x^3 {}^9C_1 x + {}^4C_4 x^4$$

$\therefore$  coefficient of  $x^4$  is

$${}^4C_0 {}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + {}^4C_4 {}^9C_0$$

$$\text{since } (1+x)^4(1+x)^9 = (1+x)^{13}$$

$${}^4C_0 {}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + {}^4C_4 {}^9C_0 = {}^{13}C_4$$

$$b) {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n = (1+x)^n$$

$$\text{sub } x=1$$

$$1 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_{n-1} + 1 = 2^n$$

$${}^nC_1 + {}^nC_2 + \dots + {}^nC_{n-1} = 2^n - 2$$

$$c) 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n$$

$$i) \text{ sub } x = -1$$

$$1 - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + (-1)^n \binom{n}{n} = 0$$

ii) differentiate both sides:

$$\binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \dots + n\binom{n}{n}x^{n-1} = n(1+x)^{n-1}$$

$$\text{sub } x=1$$

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} = n \cdot (2)^{n-1}$$

iii) Integrate both sides

$$x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \dots + \frac{1}{n+1}\binom{n}{n}x^{n+1} = \frac{(1+x)^{n+1}}{n+1} + C$$

to find  $C$ , let  $x=0$

$$0 = \frac{(1+0)^{n+1}}{n+1} + C$$

$$\therefore C = \frac{-1}{n+1}$$

$$\therefore x + \frac{1}{2} \binom{n}{1} x^2 + \frac{1}{3} \binom{n}{2} x^3 + \dots + \frac{1}{n+1} \binom{n}{n} x^{n+1} = \frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1}$$

when  $x = -1$

$$-1 + \frac{1}{2} \binom{n}{1} - \frac{1}{3} \binom{n}{2} + \dots + (-1)^{n+1} \frac{1}{n+1} \binom{n}{n} = -\frac{1}{n+1}$$

$$-1 + \frac{1}{2} \binom{n}{1} - \frac{1}{3} \binom{n}{2} + \dots + (-1)^n \cdot (-1)^1 \frac{1}{n+1} \binom{n}{n} = -\frac{1}{n+1}$$

$$1 - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} - \dots + (-1)^n \frac{1}{n+1} \binom{n}{n} = \frac{1}{n+1}$$